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Gaussian process-based prognostics of lithium-ion batteries and design optimization of cathode active materials

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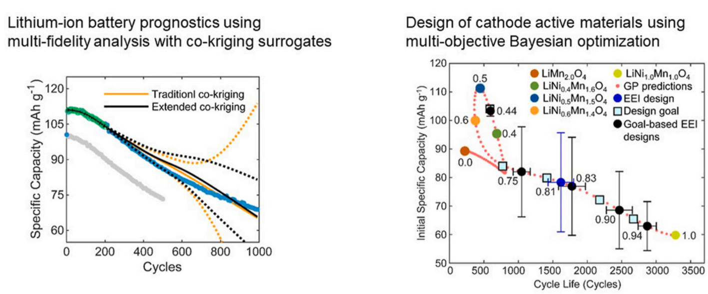
HIGHLIGHTS

- Expensive data is enhanced with cheap data for battery prognostics.
- Extended co-kriging surrogates capture the capacity degradation profile of LIB cells.
- Cell cathode materials are optimized for specific capacity and cycle life.
- A goal-based acquisition function enables parallel Bayesian optimization.

GRAPHICAL ABSTRACT



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ABSTRACT

The increasing adoption of lithium-ion batteries (LIBs) in consumer electronics, electric vehicles, and smart grids poses two challenges: the accurate prediction of the battery health to prevent operational impairments and the development of new materials for high-performance LIBs. Characterized by their flexibility and mathematical tractability, Gaussian processes (GPs) provide a powerful framework for monitoring and optimization tasks. This study employs two GP-based techniques: co-kriging surrogate modelling and Bayesian optimization. The GP training data comes from the cycling performance test of five CR2032 cells with Ni contents of 0.0, 0.4, 0.5, 0.6, and 1.0 in their cathode active material $LiNi_xMn_{2-x}O_4$. The co-kriging surrogate predicts the capacity degradation profile of a cell by exploiting information from different cells. Bayesian optimization identifies new Ni compositions that can produce cells with high initial specific capacity and large cycle life. The study shows the predictive capabilities of the co-kriging surrogate when correlated data is available. Bayesian optimization predicts that a Ni content of 0.44 produces cells with an initial specific capacity of 103.4 ± 3.8 mAh g^{-1} and a cycle life of 595 ± 12 cycles. Furthermore, the Bayesian strategy identifies other Ni contents that can improve the overall quality of the current Pareto front.

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1. Introduction

In recent years, lithium-ion batteries (LIBs) have become the most popular energy storage technology in many applications including portable electronic devices, handset tools, electric vehicles, and smart grids. In comparison to other rechargeable technologies, e.g., lead-acid, Ni-Cd, and Ni-MH, LIBs offer higher specific energy density, higher volumetric energy density, longer lifetime, lower self-discharge rate, and faster charging time [1,2]. The increasing adoption of LIBs in industrial applications requires the development of strategies to predict the degradation of LIBs in order to prevent operational impairments and catastrophic failures, and to design new materials for high-performance LIB cells. This study presents the co-kriging surrogate and Bayesian optimization as strategies that can be used in monitoring and design of LIBs.

1.1. Surrogate modeling

In general, there are three types of models to predict the degradation of LIBs: electrochemical models, equivalent circuit-based models, and data-driven models. Electrochemical models use knowledge about the material chemistry and loading conditions to analyze the loss of lithium ions and active materials. These models provide detailed explanations of the degradation mechanisms at the material level but are not reliable at larger scales due to the complex multi-scale interactions in a LIB [3,4]. Equivalent circuit-based models utilize several electric components to capture internal features of the battery, e.g., the capacitance between the electrolyte and solid material or the solid and liquid phase diffusion properties [5]. The parameters of these models are identified using large training datasets coming from battery degradation experiments. Data-driven models use past experimental data, e.g., capacity, voltage, current, temperature, etc., to predict the degradation of the battery without the need for in-depth knowledge about the battery aging mechanisms [4,6]. Therefore, data-driven models have gradually become the main methodology for battery prognostics due to their easy-to-use and flexible modeling properties [7,8].

Among the wide variety of data-driven models, statistical approaches are a popular choice in battery prognostics because they provide both predictions of the battery state and quantification of the uncertainty in the prediction, which can be used in the development of robust battery management strategies [8–12]. Examples of statistical methods that have been utilized in battery prognostics include the relevance vector machine (RVM) [8,13–15], the autoregressive (AR) model [9–12,16], and several GP-based models [17–22].

The RVM employs Bayesian learning theory to produce sparse regression models with fast testing performance [8,23]. RVMs have supported the derivation of empirical models for capacity fade prognostics and cycle life prediction [13]. Furthermore, they have been used in the identification of the parameters of particle filter (PF) algorithms for the remaining useful life (RUL) estimation of LIBs [14,15]. The main limitations of RVMs in battery prognostics are their long training times and their poor performance in extrapolation [8,23].

The AR model is a popular statistical time series approach to predict near-future values in a series [9]. In battery prognostics, the AR model has been applied in the simulation of capacity degradation profiles and the prediction of battery health indicators such as the state of health (SOH) or the RUL [9]. The prediction of a standard AR model is a linear combination of previous states. The order of determination, i.e., the number of previous states, dictates the performance of the AR model. Optimization strategies such as particle swarm optimization have been used to search for the optimal AR order [9]. An enhanced version of the standard AR approach is the nonlinear degradation autoregressive (NDAR) model [10]. The NDAR model employs an accelerated degradation factor to capture the nonlinear behavior of the capacity degradation process.

Due to its ability to model nonlinear and non-Gaussian systems, the PF technique has been widely adopted in the prediction of the capacity degradation of LIBs and RUL [11,12]. Two aspects that determine the performance of PF approaches are the choice of the importance function and the diversity of the samples. The use of the unscented Kalman filter to generate a proposal distribution as the importance function and the linear optimizing combination resampling (LOCR) algorithm to promote particle diversity has led to satisfactory results [11]. Other extensions of the standard AR approach in battery prognostics employ two functions, a linear state function and a non-linear measurement function [12]. This method requires less computational effort than the NDAR model since the linear state function enables the use of Gaussian quadrature instead of the thousands of samples required by the PF algorithm.

Another data-driven model with the ability to provide probabilistic predictions is the Gaussian process (GP) regression model [17–22]. Some of the benefits of GP regression models are their flexibility and mathematical tractability. They provide closed-form expressions of their posterior predictive distributions. GP regression models with additive covariance structures have been used to predict the capacity fade profiles of LIBs and SOH [20,21]. A popular covariance structure includes two components, a squared-exponential component and a periodic component, to capture the global degradation and local regeneration of the battery capacity, respectively [20]. A Gaussian noise term can be added to the covariance function; however, such formulation can overestimate the experimental noise generating wide confidence intervals [21]. Other alternatives to improve the predictive performance of GP models are the use of explicit mean functions and multi-output GPs [19]. Explicit mean functions and multi-output GPs regression models can be used to capture the physics of the degradation process and exploit data from multiple cells, respectively [19]. However, these strategies incur a large computational cost due to the large number of hyperparameters to be estimated and the inversion of fully coupled multi-output covariance matrices [19].

In addition to the cycle number, the degradation phenomenon can be predicted using other features linked to the aging mechanisms of LIBs. For example, several informative features can be extracted from charging curves to generate SOH predictions even under arbitrary loading profiles [22]. Other alternatives to generate predictive features include increment capacity analysis (ICA), differential voltage analysis (DVA), and differential thermal voltammetry (DTV) [18]. Recently, a matrix-variate deep Gaussian process regression (MVDGPR) model simultaneously performs three prognostics tasks: SOH estimation, end-of-life prediction, and degradation mode diagnosis [17]. The deep architecture of the MVDGPR model enabled the utilization of raw data coming from partial charge-discharge time-series related to voltage, temperature, and current. The MVDGPR model generated satisfactory predictions at different C-rates, operational temperatures, and usage profiles. The main limitation of the MVDGPR model is its computational cost. The use sampling techniques is required since deep GP architectures do not have closed-form probability distributions, which results in computationally demanding training and predictive processes.

The limitations of current GP architectures can be mitigated by the use of multi-fidelity modeling, i.e., enhancement of high-fidelity expensive data with low-fidelity cheap data. One of the most effective GP-based methods to analyze multi-fidelity data is the use of co-kriging surrogates [24–26]. The first part of this study proposes the use of the co-kriging surrogate to predict the capacity degradation profiles of LIBs. Multi-fidelity modeling allows the integration of several sources that differ in their accuracy and evaluation cost. In this manner, abundant observations from simple sources can be combined with a few observations of accurate, expensive sources to produce an enhanced predictive model [27,28]. In comparison to other data-driven models, the co-kriging surrogate offers three advantages: closed-form predictive distributions, exploitation of several sources of information, and reduced computational cost. The co-kriging surrogate employs a weighted sum of latent GPs to produce valid covariance functions and closed-form predictive equations. The co-kriging covariance functions generate positive semi-definite covariance matrices that capture two dependencies: dependencies between observations from a specific source and dependencies across observations from different sources [24, 29]. The co-kriging surrogate uses a conditional independency assumption that alleviates the computational cost of the training process since training a co-kriging surrogate is equivalent to training several GP regression models in separate [24,29,30]. Recently, a recursive scheme was derived to make computationally efficient co-kriging predictions [26]. Instead of inverting a fully coupled co-kriging covariance matrix, the recursive scheme performs inversions of several small covariance matrices.

In the prognostics of LIBs, the co-kriging surrogate can integrate data that shares some relationship, e.g., the degradation profiles of cells with slightly different cathode active materials but subjected to the same cycling protocol, or cells with the same chemistry but subjected to slightly different cycling protocols. This study employs the information from the first cycles of the cycling performance test of a LIB cell and the complete cycling information of a related cell to predict the capacity of the remaining cycles. The performance of the co-kriging surrogate is tested and compared with a multi-output GP regression model [19] using three datasets. The first dataset comes from the cycling performance test of five CR2032 cells with $LiNi_xMn_{2-x}O_4$ cathodes, where x is the content of Ni in the cathode active material. The last two datasets are taken from the open-source repository of the NASA Ames Prognostics Center of Excellence [31,32]. The results show that the performance of these two GP-based models depends on the degree of similarity between the sources. Both models have accurate predictions when the sources have a high degree of similarity. In the case of intermediate and low similarity, the multi-output GP model tends to overfit the data whereas the co-kriging surrogate displays a robust behavior.

1.2. Design optimization

In addition to battery prognostics, a potential area of application of GP-based methods is the design of cathode active materials for LIBs. Essentially all of the currently commercialized LIBs have graphitic anodes; therefore, the cathode's active material is one of the most important factors that determine the characteristics of the battery [1,2]. Predominant cathode active materials in commercial batteries contain lithium transition metal oxides ($LiMO_2$, where M = Ni, Mn, or Co), e.g., a popular choice for portable electronic devices are cells with $LiCoO_2$ cathodes due to their high specific energy, 150–200 Wh kg^{-1} [33]. Major drawbacks of $LiCoO_2$ cathodes are a relative short life span, low thermal stability, low specific power, and high cost [1,34]. This has motivated the development of new technologies such as the $LiNi_xMn_{2-x}O_4$ cathode, where x is the content of Ni [1].

$LiNi_xMn_{2-x}O_4$ can be reduced and oxidized at high potentials in non-aqueous electrolytes and the three-dimensional layout of its cubic spinel structure facilitates the diffusion of Li ions. These characteristics have enabled the fabrication of high rate $LiNi_xMn_{2-x}O_4$ batteries that match the specific energy of their $LiCoO_2$ counterparts [35]; however, there are still several challenges that need a solution. Mn-rich phases are characterized by their tendency to dissolve into the electrolyte affecting the structural stability of the active material and producing irreversible phases and inactive interfaces [1,36]. Because of the dissolution process, $LiNi_xMn_{2-x}O_4$ batteries present limited cycling performance and reduced calendar life. Two popular strategies to improve the performance of $LiNi_xMn_{2-x}O_4$ cathodes are the use of additive materials [37, 38] and surface coatings [39,40]. These strategies are often an art and well protected by trade secrets. A more accessible alternative to produce high performance batteries is the use of optimization techniques to fine tune the content of Ni in the $LiNi_xMn_{2-x}O_4$ cathode. The second part of this study employs Bayesian optimization to find high performance Ni contents.

The design of active materials for LIBs requires the evaluation of expensive black-box functions. A black-box function is an entity for which a closed-form expression is not available; however, the function can be evaluated at any point of the design domain using numerical simulations or physical experiments. The cycling performance test, which can take several days to be completed, is an example of an expensive black-box function. An additional challenge in the design of materials for LIBs is the optimization of multiple conflicting goals, e.g., high energy density and large cycle life [41]. Currently, the development of new active materials is a trial-and-error process driven by the designer's expertise. After several iterations of data analysis and testing, promising active materials are discovered but after long development times. The use of optimization techniques has mainly focused on the design of LIB electrodes and larger systems [42,43]. Among the wide variety of optimization approaches, Bayesian optimization is a well-suited approach to support the design of new cathode active materials.

Recently, Bayesian optimization has emerged as an efficient methodology to solve optimization problems that require the evaluation of expensive black-box functions. Bayesian optimization has two main components: a probabilistic surrogate model of the black-box function and an acquisition function that guides the optimization [44,45]. Among the wide variety of options, the GP regression model is the most popular surrogate model in Bayesian optimization due to its predictive power and mathematical tractability [44]. After training, the surrogate model provides predictions of the black-box function at unseen locations and estimates of the accuracy of the predictions [46]. The probabilistic predictions of the surrogate model are used in the formulation of the acquisition function, which quantifies the gain that can be attained by sampling a design at a new location. At each iteration, the acquisition function is maximized to identify new infilling candidates. Given that Bayesian optimization operates in a probabilistic framework, it does not require gradient information; however, if available, such information can be used to enhance the surrogate model [47]. Recently, Bayesian optimization has been applied in the design of solid-state electrolytes [48–51], the design of liquid electrolytes [52], and the improvement of the open circuit voltage of LIB cathodes [53]. Our research group has obtained preliminary results in the design of cathode active materials using a constrained acquisition function [54].

This work utilizes Bayesian optimization in the design of $LiNi_xMn_{2-x}O_4$ cathodes by searching for Ni contents that can produce cells with high initial specific capacity and/or large cycle life. The training data is extracted from the cycling performance test of five CR2032 cells with Ni contents of 0.0, 0.4, 0.5, 0.6, and 1.0. The optimization strategy employs GP regression models and a goal-based multi-objective acquisition function. Contrarily to typical acquisition functions, the goal-based formulation enables the identification of several designs at a given iteration. i.e., parallel optimization. The results suggest that a Ni content of 0.44 produces cells with an initial specific capacity of 103.4 ± 1.9 mAh g^{-1} and a cycle life of 595 ± 6 cycles, and that Ni contents of 0.7, 0.8, and 0.9 can improve the overall quality of the current Pareto front.

2. Methodology

2.1. Gaussian process regression

In engineering design, surrogate models are fast mathematical approximations of functions that require long-running times or substantial resources for their evaluation, e.g., cycling performance test of LIBs. A surrogate model maps an input $\mathbf{x}$ to an output $f(\mathbf{x})$ given a set of n training samples $\mathbf{X} = \{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\}$ with output values $\mathbf{f} = \{f^{(1)}, \ldots, f^{(n)}\}$. In some modeling problems, the data does not correspond to observations of $f(\mathbf{x})$, but to noisy data $y(\mathbf{x})$. Noisy data is frequently modelled as $y(\mathbf{x}) = f(\mathbf{x}) + \varepsilon$, where ε is an independent Gaussian noise $\varepsilon \sim \mathcal{N}(0, {\sigma_n}^2)$ and σ_n^2 is the variance of the noise. Once trained, surrogate models can be used to make predictions at test locations $\mathbf{X}_*$. This study employs Gaussian process-based surrogate models with two purposes: (1) to forecast the capacity degradation profiles of LIB cells under cycling loads and (2) to predict the initial specific capacity and cycle life of $LiNi_xMn_{2-x}O_4$ cells as a function of the content of Ni in their cathode active material.

Gaussian process (GP) regression models, also known as kriging surrogates, refer to the use of GPs as probabilistic surrogate models. A GP is denoted as

$$f(\mathbf{x}) \sim \mathscr{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')), \tag{1}$$

where $m(\mathbf{x})$ is the mean function and $k(\mathbf{x}, \mathbf{x}')$ is the covariance function. In GP-based modeling, the mean and covariance functions are utilized to incorporate prior knowledge about $f(\mathbf{x})$ in the regression model [44,55,56]. For example, the use of a linear mean function and a squared-exponential covariance function leads to a surrogate model with a global linear behavior and smooth non-linear local variations.

Often, the mean function is defined as $m(\mathbf{x}) = 0$ since the covariance function $k(\mathbf{x}, \mathbf{x}')$ is flexible enough to model the data [19,55,57]. The predictive equations of a GP regression model can be obtained analytically by employing the rules for conditioning Gaussian distributions [55]. The predictive equations for a zero-mean GP regression model are

$$\mathbf{f}_{zm} | \mathbf{X}, \mathbf{y}, \mathbf{X}_* \sim \mathscr{N}\left(\overline{\mathbf{f}}_{zm}, \mathrm{cov}(\mathbf{f}_{zm})\right), \tag{2}$$

where

$$\overline{\mathbf{f}}_{zm} = \mathbf{K}(\mathbf{X}_*, \mathbf{X})\left[\mathbf{K}(\mathbf{X}, \mathbf{X}) + \sigma_n^2 \mathbf{I}\right]^{-1} \mathbf{y} \tag{3}$$

and

$$\mathrm{cov}(\mathbf{f}_{zm}) = \mathbf{K}(\mathbf{X}_*, \mathbf{X}_*) - \mathbf{K}(\mathbf{X}_*, \mathbf{X})\left[\mathbf{K}(\mathbf{X}, \mathbf{X}) + \sigma_n^2 \mathbf{I}\right]^{-1} \mathbf{K}(\mathbf{X}, \mathbf{X}_*). \tag{4}$$

Here, $\overline{\mathbf{f}}_{zm}$ is the predictive mean of the GP regression model. It predicts the response of $f(\mathbf{x})$ at the unseen locations $\mathbf{X}_*$. The diagonal of $\mathrm{cov}(\mathbf{f}_{zm})$ is the predictive variance of the GP model. It provides estimates of the uncertainty in the predictions. In these equations, $\mathbf{K}(\mathbf{X}, \mathbf{X})$ is the covariance matrix of the observed data, σ_n^2 is the variance of the noise in the observations, $\mathbf{I}$ is the $n \times n$ identity matrix, $\mathbf{K}(\mathbf{X}, \mathbf{X}_*)$ is the covariance matrix between the observed data and the predictions, and $\mathbf{K}(\mathbf{X}_*, \mathbf{X}_*)$ is the covariance matrix of the predictions. The components of the covariance matrices are generated by the evaluation of the covariance function $k(\mathbf{x}, \mathbf{x}')$.

Valid covariance functions produce positive semi-definite covariance matrices regardless of the chosen pair of points $(\mathbf{x}, \mathbf{x}')$ [45]. This study employs the following covariance function

$$k(x_p, x_q) = \sigma_f^2 \exp\left(-\frac{1}{2l^2}(x_p - x_q)^2\right) + \sigma_n^2 \delta(x_p, x_q), \tag{5}$$

where the first term is the squared-exponential covariance function, which produces smooth predictions, and the second term captures the noise in the data. The term $\delta(x_p, x_q)$ is the Kronecker delta function. The parameters σ_f^2, σ_n^2, and l are known as the hyperparameters of the GP model. Here, σ_f^2 is the variance of the noise-free signal, σ_n^2 is the variance of the noise, and l is the characteristic length of the function [55]. This work employs the maximum likelihood estimation (MLE) approach [55] to estimate the hyperparameters, i.e., train the GP regression model.

If there is prior knowledge about the behavior of the function, such knowledge can be modelled as

$$f(\mathbf{x}) = \mathbf{h}(\mathbf{x})^{\mathrm{T}} \boldsymbol{\beta} + f_{zm}(\mathbf{x}), \tag{6}$$

where $\mathbf{h}(\mathbf{x})$ is a set of basis functions, $\boldsymbol{\beta}$ are weighting coefficients with a prior distribution $\boldsymbol{\beta} \sim \mathscr{N}(\mathbf{b}, \mathbf{B})$, and $f_{zm}(\mathbf{x})$ is a GP with a zero mean function and covariance function $k(\mathbf{x}, \mathbf{x}')$ [55]. Here, $\mathbf{h}(\mathbf{x})^{\mathrm{T}} \boldsymbol{\beta}$ models the expected behavior of the function and $f_{zm}(\mathbf{x})$ accounts for unknown patterns. The resulting GP is

$$f(\mathbf{x}) \sim \mathscr{GP}\left(\mathbf{h}(\mathbf{x})^{\mathrm{T}} \mathbf{b}, k(\mathbf{x}, \mathbf{x}') + \mathbf{h}(\mathbf{x})^{\mathrm{T}} \mathbf{B} \mathbf{h}(\mathbf{x}')\right). \tag{7}$$

Equation (7) shows the effect of the basis functions and the prior of the weighting coefficients in the mean and covariance functions of the GP model.

2.2. Multi-fidelity analysis using Co-kriging surrogates

Many modeling scenarios have multiple sources of information that differ in their level of accuracy and acquisition cost. A co-kriging surrogate enables the integration of a few observations from an expensive but accurate source with a large number of observations from fast to evaluate but less accurate sources. By combining this information, the co-kriging approach builds a regression model with better predictive capabilities than a surrogate model that only uses the expensive observations.

Traditional co-kriging formulations use GPs with constant mean functions and non-informative priors [24,29,58], which offers mathematical simplicity in the derivation of the predictive equations. This paper presents the formulation of a co-kriging surrogate that employs informative basis functions in order to exploit prior knowledge regarding the degradation of LIBs. The surrogate model combines two sources: a high-fidelity, expensive to evaluate source with n_e noisy values $\mathbf{y}_e = \{y_e^{(1)}, \ldots, y_e^{(n_e)}\}$ at locations $\mathbf{X}_e = \{\mathbf{x}_e^{(1)}, \ldots, \mathbf{x}_e^{(n_e)}\}$ and a low-fidelity, fast to evaluate source with n_c noisy values $\mathbf{y}_c = \{y_c^{(1)}, \ldots, y_c^{(n_c)}\}$ at locations $\mathbf{X}_c = \{\mathbf{x}_c^{(1)}, \ldots, \mathbf{x}_c^{(n_c)}\}$. In this study, the expensive source is the capacity degradation profile of a LIB cell under cycling load and the low-fidelity source is the capacity degradation profile of a related cell. The co-kriging model forecasts the degradation of the cell through the cycling experiment.

The high-fidelity source $y_e(\mathbf{x})$ is modelled as the weighted sum of two independent GPs

$$y_e(\mathbf{x}) = \rho y_c(\mathbf{x}) + y_d(\mathbf{x}), \tag{8}$$

where $y_c(\mathbf{x})$ is a GP model of the low-fidelity source, $y_d(\mathbf{x})$ is a corrective GP model that captures the difference between $y_e(\mathbf{x})$ and $\rho y_c(\mathbf{x})$, and ρ is a scaling factor. This study employs $y_c(\mathbf{x})$ and $y_d(\mathbf{x})$ of the form

$$y_c(\mathbf{x}) \sim \mathscr{GP}\left(\mathbf{h}_c(\mathbf{x})^{\mathrm{T}} \mathbf{b}_c, k_c(\mathbf{x}, \mathbf{x}') + \mathbf{h}_c(\mathbf{x})^{\mathrm{T}} \mathbf{B}_c \mathbf{h}_c(\mathbf{x}')\right) \tag{9}$$

and

$$y_d(\mathbf{x}) \sim \mathscr{GP}\left(\mathbf{h}_d(\mathbf{x})^{\mathrm{T}} \mathbf{b}_d, k_d(\mathbf{x}, \mathbf{x}') + \mathbf{h}_d(\mathbf{x})^{\mathrm{T}} \mathbf{B}_d \mathbf{h}_d(\mathbf{x}')\right), \tag{10}$$

where $\mathbf{h}_c(\mathbf{x})$ and $\mathbf{h}_d(\mathbf{x})$ are the two sets of basis functions. The weighting coefficients of the basis functions of $y_c(\mathbf{x})$ and $y_d(\mathbf{x})$ have prior distributions $\boldsymbol{\beta}_c \sim \mathscr{N}(\mathbf{b}_c, \mathbf{B}_c)$ and $\boldsymbol{\beta}_d \sim \mathscr{N}(\mathbf{b}_d, \mathbf{B}_d)$, respectively. These basis functions and prior distributions capture the knowledge that the designer has about the low-fidelity source and corrective model.

Since Equation (8) corresponds to a weighted sum of two GPs, $y_e(\mathbf{x})$ is another GP due to the closure property of GPs under linear operations [47,59]. Therefore, it is possible to derive a closed-form of the predictive equations of the co-kriging model employing the conditioning rules of Gaussian distributions [55]. The predictive equations of the high-fidelity source are

$$\overline{\mathbf{f}}_p = \mathbf{m}_e(\mathbf{X}_*) + \mathbf{C}^{\mathrm{T}} \mathbf{D}^{-1} \left(\begin{bmatrix} \mathbf{y}_c \\ \mathbf{y}_e \end{bmatrix} - \begin{bmatrix} \mathbf{m}_c(\mathbf{X}_c) \\ \mathbf{m}_e(\mathbf{X}_e) \end{bmatrix} \right) \tag{11}$$

and

$$\mathrm{cov}(\mathbf{f}_p) = \mathbf{K}_{pp}(\mathbf{X}_*, \mathbf{X}_*) - \mathbf{C}^{\mathrm{T}} \mathbf{D}^{-1} \mathbf{C}, \tag{12}$$

where

$$\mathbf{D} = \begin{bmatrix} \mathbf{K}_{cc}(\mathbf{X}_c, \mathbf{X}_c) & \mathbf{K}_{ce}(\mathbf{X}_c, \mathbf{X}_e) \\ \mathbf{K}_{ce}(\mathbf{X}_e, \mathbf{X}_c) & \mathbf{K}_{ee}(\mathbf{X}_e, \mathbf{X}_e) \end{bmatrix} \tag{13}$$

and

$$\mathbf{C} = \begin{bmatrix} \mathbf{K}_{cp}(\mathbf{X}_c, \mathbf{X}_*) \\ \mathbf{K}_{ep}(\mathbf{X}_e, \mathbf{X}_*) \end{bmatrix}. \tag{14}$$

Here, $\overline{\mathbf{f}}_p$ is the predictive mean of the co-kriging surrogate. It predicts the response of the high-fidelity source at the unseen locations $\mathbf{X}_*$, e.g., the capacity of the LIB cell during the last cycles of the cycling test. The diagonal of $\mathrm{cov}(\mathbf{f}_p)$ is the predictive variance of the co-kriging model. In these equations, $\mathbf{D}$ is a covariance matrix that captures dependencies of the high-fidelity data $\mathbf{K}_{ee}(\mathbf{X}_e, \mathbf{X}_e)$ and of the low-fidelity data $\mathbf{K}_{cc}(\mathbf{X}_c, \mathbf{X}_c)$, and the cross-covariance between the high- and low-fidelity data $\mathbf{K}_{ce}(\mathbf{X}_c, \mathbf{X}_e)$. $\mathbf{C}$ is a matrix composed by the covariance between the low-fidelity data and the predictions $\mathbf{K}_{cp}(\mathbf{X}_c, \mathbf{X}_*)$, and the covariance between the high-fidelity data and the predictions $\mathbf{K}_{ep}(\mathbf{X}_e, \mathbf{X}_*)$. Furthermore, $\mathbf{K}_{pp}(\mathbf{X}_*, \mathbf{X}_*)$ is covariance matrix of the predictions. The vectors $\mathbf{m}_e(\mathbf{X}_*)$, $\mathbf{m}_e(\mathbf{X}_e)$, and $\mathbf{m}_c(\mathbf{X}_c)$ are generated by the mean functions of the co-kriging model. The Supplementary Information contains a detailed derivation of the co-kriging surrogate and the mean and covariance functions to calculate $\overline{\mathbf{f}}_p$ and $\mathrm{cov}(\mathbf{f}_p)$.

The inference of the hyperparameters of a co-kriging surrogate is supported by the following conditional independence assumption

$$\mathrm{cov}\{y_e(\mathbf{x}), y_c(\mathbf{x}')|y_c(\mathbf{x})\} = 0,\ \forall \mathbf{x} \neq \mathbf{x}'. \tag{15}$$

Equation (15) implies that if the value of $y_c(\mathbf{x})$ is known at the location $\mathbf{x}$, no more can be learned about the high fidelity source $y_e(\mathbf{x})$ at the same location $\mathbf{x}$ from additional observations $y_c(\mathbf{x}')$ at other locations $\mathbf{x}'$ [24,25,29]. Consequently, the GP models of $y_c(\mathbf{x})$ and $y_d(\mathbf{x})$ can be trained independently. The hyperparameters of $y_c(\mathbf{x})$ are determined using the data $(\mathbf{X}_c,\ \mathbf{y}_c)$. The data to find the hyperparameters of $y_d(\mathbf{x})$ corresponds to the difference between observations of the high fidelity source $\mathbf{y}_e$ and $\rho y_c(\mathbf{x})$ only at the locations $\mathbf{X}_e$, e.g., $y_d{}^{(1)} = y_e{}^{(1)} - \rho y_c(\mathbf{x}_e{}^{(1)})$. This definition assumes that $\mathbf{X}_e$ is a subset of $\mathbf{X}_c$, i.e., $\mathbf{X}_e \subseteq \mathbf{X}_c$. If the locations do not coincide, the GP regression model of the low fidelity source can be used to estimate the values of $y_c(\mathbf{x})$ at the non-collocated locations [25]. This study employs the MLE approach to train the co-kriging surrogate [55]. The genetic algorithm implementation of Matlab is utilized to find the hyperparameters that maximize the log marginal likelihood of $y_c(\mathbf{x})$ and $y_d(\mathbf{x})$.

In general, the scaling factor ρ is estimated along with the hyperparameters of the second GP model $y_d(\mathbf{x})$. This approach produces satisfactory results for interpolation. In extrapolation, e.g., forecasting of the capacity degradation profiles of LIBs, satisfactory predictions are obtained by using a scaling factor $\rho = 1$. This assumption implies that the low fidelity source $y_c(\mathbf{x})$ describes an expected degradation profile of $y_e(\mathbf{x})$, and that $y_d(\mathbf{x})$ corrects deviations from such expectation. Another alternative to model ρ is the use of a prior distribution that captures the prior knowledge about the similarity between the sources; however, such formulation leads to a model of $y_e(\mathbf{x})$ that is no longer a GP, which prevents the derivation of closed-form predictive equations.

2.3. Multi-objective Bayesian optimization

In Bayesian optimization, the expected improvement (EI) is the most popular acquisition function to solve single-objective optimization problems. This function estimates the gain that a new sample $\mathbf{x}$ will produce in the current sampling plan. For a minimization problem, the EI function is defined as

$$\mathrm{E}[I(\mathbf{x})] = \mathrm{E}[\max(f_{\min} - f(\mathbf{x}), 0)], \tag{16}$$

where $f_{\min} = \min\{f^{(1)}, \dots, f^{(n)}\}$ is the best current design. Commonly, $f(\mathbf{x})$ is a GP regression model with a predictive mean $\overline{f}(\mathbf{x})$ and predictive variance $\mathrm{cov}(f(\mathbf{x}))$, which leads to the following closed-form expression for the EI function

$$\mathrm{E}[I(\mathbf{x})] = (f_{\min} - \overline{f}(\mathbf{x}))\Phi(u(\mathbf{x})) + s(\mathbf{x})\varphi(u(\mathbf{x})), \tag{17}$$

where $u(\mathbf{x}) = (f_{\min} - \overline{f}(\mathbf{x}))/s(\mathbf{x})$ and $s(\mathbf{x}) = \sqrt{\mathrm{cov}(f(\mathbf{x}))}$. In here, $\varphi(\cdot)$ and $\Phi(\cdot)$ are the probability density function and the cumulative density function for a standard Gaussian distribution [25].

During the solution of the optimization problem, the EI function is maximized to determine the next design to be sampled or to finalize the optimization. If the maximum EI is larger than a threshold value, the new design is sampled and the GP regression model is updated; otherwise, the optimization ends (Fig. 1 (a)). Maximization of the EI function balances exploration of the design space and exploitation of promising design zones [25,44,57].

In the case of multi-objective problems, EI-based acquisition functions quantify the gain over the current Pareto front offered by a new design $\mathbf{x}$. One example of such a formulation is the Euclidean distance-based expected improvement (EEI) [60]. For a minimization problem with two objectives functions $f_1(\mathbf{x})$ and $f_2(\mathbf{x})$, the evaluation of the objectives using a sampling plan $\mathbf{X}$ produces an initial approximation of the Pareto front

$$\mathbf{f}^*_{1,2} = \left\{ \left(f_1^{*(1)}, f_2^{*(1)}\right),\ \dots,\ \left(f_1^{*(m)}, f_2^{*(m)}\right) \right\}, \tag{18}$$

where $f_j^{*(i)} = f_j(\mathbf{x}^{*(i)})$, $\mathbf{x}^{*(i)}$ is the i-th design of the current Pareto front, and m is the number of designs in the current Pareto front. The EEI function is defined as

$$\mathrm{E}[I(\mathbf{x})] = \mathrm{P}[I(\mathbf{x})]d_{\min}, \tag{19}$$

where $\mathrm{P}[I(\mathbf{x})]$ is the probability of improving both functions $f_1(\mathbf{x})$ and $f_2(\mathbf{x})$ if the design $\mathbf{x}$ is sampled, and $d_{\min}$ is the Euclidean distance between the centroid of the probability of improvement $\mathrm{P}[I(\mathbf{x})]$ and the closest design in the current Pareto front (f_1^*, f_2^*) (Fig. 1 (b)). When using GP regression models for $f_1(\mathbf{x})$ and $f_2(\mathbf{x})$, the probability of improvement is

$$\mathrm{P}[I(\mathbf{x})] = \Phi\left(u_1^1\right) + \sum_{i=1}^{m-1}\left[\Phi\left(u_1^{i+1}\right) - \Phi\left(u_1^i\right)\right]\Phi\left(u_2^{i+1}\right) + \left[1 - \Phi\left(u_1^m\right)\right]\Phi\left(u_2^m\right), \tag{20}$$

where $u_j^i = (f_j^{*(i)} - \overline{f}_j(\mathbf{x}))/s_j(\mathbf{x})$. The centroid of the probability of improvement $(\overline{F}_1(\mathbf{x}), \overline{F}_2(\mathbf{x}))$ is given by

$$\overline{F}_1(\mathbf{x}) = \left[z_1^1 + \sum_{i=1}^{m-1}\left(z_1^{i+1} - z_1^i\right)\Phi\left(u_2^{i+1}\right) + z_1^m\Phi\left(u_2^m\right)\right] \Big/ \mathrm{P}[I(\mathbf{x})], \tag{21}$$

where $z_j^i = \overline{f}_j(\mathbf{x})\Phi(u_j^i) - s_j(\mathbf{x})\varphi(u_j^i)$. $\overline{F}_2(\mathbf{x})$ is defined similarly. As in the single objective case, the EEI function is maximized to determine the next design to be sampled or to end the optimization.

Traditional Bayesian optimization strategies assume that the objective functions are evaluated sequentially, i.e., only one design is sampled at each iteration [61]. However, in many design problems, it is more practical to evaluate multiple designs in parallel in order to accelerate the design process. This study employs the goal-based approach proposed in Ref. [62] that extends the EEI function to identify multiple designs at a given iteration.

The goal-based approach utilizes a prescribed goal point $\mathbf{f}_g$ to focus the attention of the optimizer over a specific region of the Pareto front. The goal-based formulation enables parallel optimization since several goals can be prescribed at a given iteration.

For a minimization problem with two objectives, the goal-based EEI function is defined as

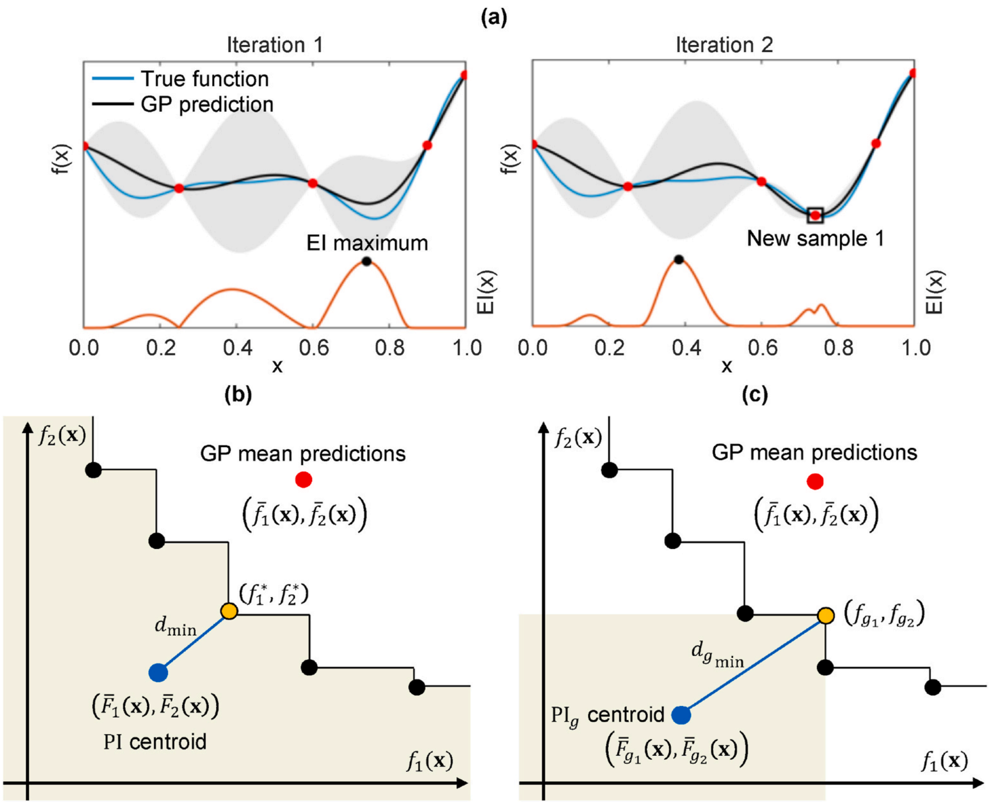

Fig. 1. Bayesian optimization. (a) Solution of a single-objective optimization problem (minimization) utilizing the EI as the acquisition function. The expected improvement is high where the mean prediction of the GP model predicts low values of $f(\mathbf{x})$ and where the uncertainty (gray area) of the predictions is high. The bottom figures show the elements of the Euclidean distance-based expected improvement (EEI) function **(b)** and the goal-based EEI function **(c)**. The black dots denote the current Pareto front. The design $\mathbf{x}$ has GP mean predictions $\bar{f}_1(\mathbf{x})$ and $\bar{f}_2(\mathbf{x})$ (red dot). The blue dot is the centroid of the probability of improvement of the design $\mathbf{x}$. In the EEI formulation, the orange dot is the closest Pareto design. In the goal-based formulation, the orange dot is the prescribed goal that defines the region of interest in the objective space (gray area). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

$$E\left[I_g(\mathbf{x})\right] = P\left[I_g(\mathbf{x})\right]\sqrt{\left(\bar{F}_{g_1}(\mathbf{x}) - f_{g_1}\right)^2 + \left(\bar{F}_{g_2}(\mathbf{x}) - f_{g_2}\right)^2}, \tag{22}$$

where $P[I_g(\mathbf{x})]$ is the probability of improving $f_1(\mathbf{x})$ and $f_2(\mathbf{x})$ over the prescribed goal $\mathbf{f}_g = \left[f_{g_1} \quad f_{g_2}\right]^T$ if the design $\mathbf{x}$ is sampled. Here, $\bar{F}_{g_1}(\mathbf{x})$ and $\bar{F}_{g_2}(\mathbf{x})$ are the coordinates of the centroid of the goal-based probability of improvement (Fig. 1 (c)).

If the surrogate models are GPs, the goal-based probability of improvement is

$$P\left[I_g(\mathbf{x})\right] = \Phi(u_1)\Phi(u_2), \tag{23}$$

where $u_1 = (f_{g_1} - \bar{f}_1(\mathbf{x}))/s_1(\mathbf{x})$ and $u_2 = (f_{g_2} - \bar{f}_2(\mathbf{x}))/s_2(\mathbf{x})$.

The centroid of the goal-based probability of improvement is given by

$$\bar{F}_{g_1}(\mathbf{x}) = \left[(\bar{f}_1(\mathbf{x})\Phi(u_1) - s_1(\mathbf{x})\varphi(u_1))\Phi(u_2)\right] / P\left[I_g(\mathbf{x})\right]. \tag{24}$$

Here, $\bar{F}_{g_2}(\mathbf{x})$ is defined similarly. If the prescribed goals lie close to the ends of the Pareto front, the expressions for the goal-based EEI function need to be modified following the procedure indicated in Ref. [62]. This study employs the genetic algorithm implementation of Matlab to maximize the EEI function and to maximize the goal-based EEI function for different prescribed goals.

2.4. Battery datasets

This study employs three datasets, which are summarized in Fig. 2. The first dataset comes from the cycling performance test of five CR2032 cells with $LiNi_xMn_{2-x}O_4$ cathodes. The cells have different contents of Ni in their cathode active material. The purpose of this dataset is twofold: (1) to study the performance of the proposed co-kriging formulation, and (2) to design $LiNi_xMn_{2-x}O_4$ cathodes. The second and third datasets are taken from the open-source repository of

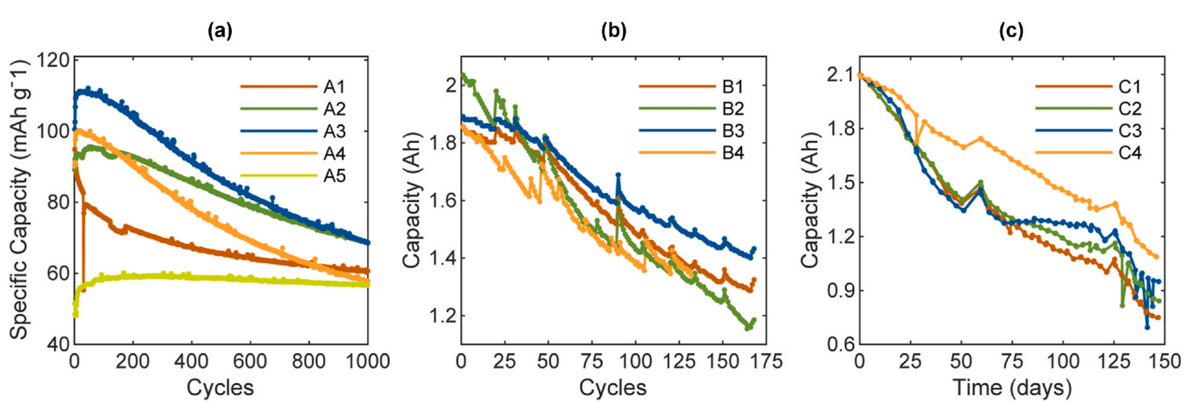

Fig. 2. Battery datasets. (a) Degradation profiles of CR2032 cells with different content of Ni in their $LiNi_xMn_{2-x}O_4$ cathode active material. **(b)** Four degradation profiles from the NASA Battery Data Set. **(c)** Four degradation profiles from the NASA Randomized Battery Usage Data Set.

the NASA Ames Prognostics Center of Excellence [31,32]. These two datasets are utilized to further study the performance of the co-kriging surrogate model.

In the first dataset, the cycling performance test of five CR2032 cells with cathode active material $LiNi_xMn_{2-x}O_4$ is performed. The cells have Ni contents of 0.0, 0.4, 0.5, 0.6, and 1.0. In this study, these cells are labeled as A1, A2, A3, A4, and A5. The $LiNi_xMn_{2-x}O_4$ electrodes are made of a 50:30:20 (weight %) mixture of active material, carbon black, and PVDF (Polyvinylidene fluoride) binder. The Supplementary Information contains further information about the preparation of the CR2032 cells. The five cells are charged and discharged with a constant current of 1 C in the voltage between 3.5 and 4.9 V during 1000 cycles. Fig. 2 (a) shows the capacity degradation profiles of the cells. This dataset is utilized to test the performance of the proposed co-kriging formulation in the forecasting of the capacity degradation of LIBs and in the identification of Ni compositions that can produce $LiNi_xMn_{2-x}O_4$ cells with high initial specific capacity and large cycle life.

The second dataset is obtained from the NASA Ames Prognostics Center of Excellence Battery Data Set [31]. It corresponds to the cycling performance test of four commercial 18650 batteries (capacity 2 Ah). The batteries are tested at room temperature and subjected to repeated charge and discharge cycles, and impedance analysis. The charging phase consists of a constant-current, constant-voltage regime. The batteries are charged with a constant current of 1.5 An until they reach an upper voltage limit of 4.2 V. Then, a constant voltage of 4.2 V is applied until the current drops to 20 mA. In the discharging phase, a constant current of 2 A is utilized until the voltage drops to 2.7 V, 2.5 V, 2.2 V, and 2.5 V for batteries 5, 6, 7, and 18, respectively. The cycling experiment stops when the batteries lost 30% of their initial capacity. Additional details of the experiment are available in Ref. [63]. In this work, the batteries 5, 6, 7, and 18 in the numbering of the online repository are labeled as B1, B2, B3, and B4 (Fig. 2 (b)). This dataset offers the opportunity to test the proposed co-kriging formulation in the presence of highly noisy data.

The third dataset is obtained from the NASA Ames Prognostics Center of Excellence Randomized Battery Usage Data Set [32]. It corresponds to four commercial 18650 batteries that are subjected to randomized current loads in order to mimic a practical battery usage. The randomized loads range from 0.5 A to 4 A. In order to represent two types of users: a mild and an aggressive user, the random walk is biased towards low and high current loads, respectively. After every fifty randomized discharging cycles, a low-rate charge-discharge protocol is utilized to obtain a standardized measure of the capacity of the batteries. A detailed description of the experiments is available in Ref. [64]. This study employs the batteries W9, W10, W11, and W12, which offer an opportunity to test the co-kriging formulation in front of highly non-linear and noisy degradation profiles (Fig. 2 (c)). For consistency, these batteries are labeled as C1, C2, C3, and C4.

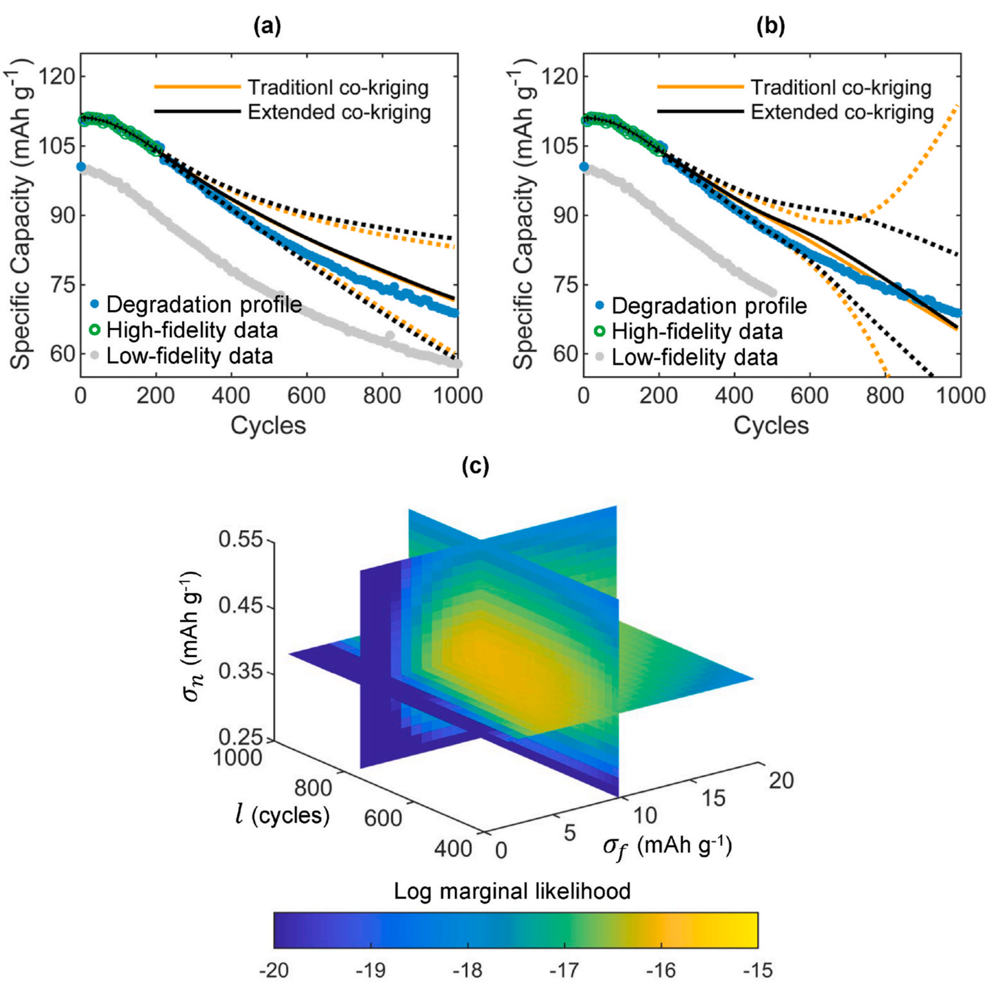

Fig. 3. Co-kriging surrogates. Predictions of a traditional co-kriging model and the extended formulation within the range of the low-fidelity source **(a)** and beyond the range of the low-fidelity source **(b)**. The dotted lines denote the 95% confidence interval. **(c)** Log marginal likelihood of the difference GP model $y_d(\mathbf{x})$ for the dataset in **(a)**.

3. Results and discussion

3.1. Prediction of the capacity degradation of LIBs

This section employs the proposed co-kriging surrogate to predict the capacity fade of a LIB cell under cycling loading. For the predictions, the co-kriging model employs two sources of information: a high-fidelity source and a low-fidelity source. The high-fidelity information is the capacity degradation profile of a cell, referred as the primary cell, up to a specific cycle. The low-fidelity information is the capacity degradation profile of a related cell, referred as the auxiliary cell, through the entire cycling test. The co-kriging surrogate employs the low- and high-fidelity data to forecast the degradation of the primary cell.

Fig. 3 illustrates some of the differences between the performance of a traditional co-kriging formulation, which employs GPs with constant mean functions, and the performance of the extended formulation, which employs GPs with informative mean functions. This comparison is generated using the cell A3 as the high-fidelity source and the cell A4 as the low-fidelity source (Fig. 2 (a)). Both co-kriging models present similar behavior through the range of the low-fidelity data (Fig. 3 (a)). Beyond this range, the models have different estimations of uncertainty (Fig. 3 (b)). The extended formulation present better estimations since it produces a tighter confidence interval that encloses the true degradation profile.

In this study, the log marginal likelihood of the training data is maximized to train the GP models of the low-fidelity source $y_c(\mathbf{x})$ and the corrective GP model $y_d(\mathbf{x})$. Fig. 3 (c) shows a contour plot of the log marginal likelihood of the corrective GP model $y_d(\mathbf{x})$ for the dataset in Fig. 3 (a). The hyperparameters of the GP model are the ones that maximize the likelihood function ($\sigma_f = 9.82$ mAh g^{-1}, $\sigma_n = 0.38$ mAh g^{-1}, and $l = 792.8$ cycles).

As in Ref. [19], the room-mean-squared error (RMSE) in the capacity estimation is utilized as a metric to quantify the performance of the forecasting model. The RMSE is defined as

$$\text{RMSE} = \sqrt{\frac{1}{n_t}\sum_{i=1}^{n_t}\left(\bar{f}_p(x_i) - f(x_i)\right)^2}, \tag{25}$$

where $\bar{f}_p(x_i)$ is the predicted capacity of the surrogate model at cycle x_i, $f(x_i)$ is the denoised true capacity of the battery at x_i, and n_t is the number of remaining cycles to complete the cycling test. A GP regression model with a zero-mean function and the covariance function in Eq. (5) is utilized to filter the noise in the capacity degradation profile.

The performance of the proposed co-kriging surrogate model is compared with the multi-output GP model presented in Ref. [19]. This multi-output model has a zero mean function and the following covariance function

$$k_{mo}([t_r \quad x_p], [t_s \quad x_q]) = k_t(t_r, t_s) \times k_x(x_p, x_q), \tag{26}$$

where t_i is an integer label to identify the t_i-th source and x_j is the cycle related to a given capacity measure $y_{t_i}(x_j)$ [19]. In here, $k_t(t_r, t_s)$ captures the covariance between the sources and $k_x(x_p, x_q)$ captures the covariance between measures of the capacity at different cycles. In this study, given that the co-kriging model is compared with a multi-output GP model that fuses two sources, the spherical parameterization of Equation (26) proposed in Ref. [19] is replaced by a covariance function whose $k_t(t_r, t_s)$ is a squared-exponential covariance function and $k_x(x_p, x_q)$ is the covariance function in Equation (5). This change facilitates the implementation of the multi-output GP model.

For each dataset (Fig. 2), the performance of the co-kriging surrogate and the multi-output GP model is studied using one primary cell and three different auxiliary cells. The degradation profiles of the primary and auxiliary cells present different levels of similarity. The Pearson correlation coefficient is utilized as a similarity index. The validation cases correspond to high, intermediate, and low similarity (Table 1).

Table 1
Test cases of the extended co-kriging surrogate model.

Dataset	High-fidelity source	Low-fidelity source		
		Case 1	Case 2	Case 3
A	A3	A4	A2	A1
B	B1	B3	B2	B4
C	C1	C2	C4	C3

In the test cases of dataset A, the primary cell A3 has active material $LiNi_{0.5}Mn_{1.5}O_4$. The auxiliary cells A4, A2, and A1 have active materials $LiNi_{0.6}Mn_{1.4}O_4$, $LiNi_{0.4}Mn_{1.6}O_4$, and $LiMn_2O_4$, respectively. To study the performance of the forecasting models, high-fidelity information from the first 100, 200, 300, 400, 500, 600, 700, and 800 cycles is utilized. The low-fidelity information are the specific capacity values through the entire cycling test (1000 cycle).

In order to capture the capacity fade of the auxiliary cell, the mean function of the GP model of $y_c(\mathbf{x})$ is a linear function with a prior distribution $\boldsymbol{\beta}_c \sim \mathcal{N}(\mathbf{b}_c, \mathbf{B}_c)$. The basis functions $\mathbf{h}_c(\mathbf{x})$, and the prior parameters $\mathbf{b}_c$, and $\mathbf{B}_c$ are

$$\mathbf{h}_c(\mathbf{x}) = \begin{bmatrix} 1 \\ x \end{bmatrix},\ \mathbf{b}_c = \begin{bmatrix} y_0 \\ m_c \end{bmatrix}, \quad and \quad \mathbf{B}_c = \begin{bmatrix} 4 & 0 \\ 0 & {m_c}^2 \end{bmatrix}, \tag{27}$$

where $y_0 = (\sum_{i=1}^{10} y_i)/10$, $m_c = (y_n - y_0)/x_n$, y_i is the specific capacity at the i-th cycle, and y_n is the specific capacity measured at the last cycle x_n. The narrow Gaussian distribution over $\boldsymbol{\beta}_c$ generates a mean function that captures the linear decay of $y_c(\mathbf{x})$.

In the case of the difference model $y_d(\mathbf{x})$, the trend and the form of the function are unknown. This is a proper scenario for the use of a constant mean function with a broad prior distribution $\boldsymbol{\beta}_d \sim \mathcal{N}(\mathbf{b}_d, \mathbf{B}_d)$. The mean function of $y_d(\mathbf{x})$ is composed by

$$\mathbf{h}_d(\mathbf{x}) = 1,\ \mathbf{b}_d = 0, \quad and\ \mathbf{B}_d = 100. \tag{28}$$

At the beginning of the cycling test, the experimental data has a large number of outliers that can hinder the performance of the forecasting models. For this reason, the training data is taken after the first 10 cycles. During training, the optimizer is required to find a length scale l_c of the low-fidelity source $y_c(\mathbf{x})$ larger than 100 cycles and a length scale l_d of the difference model $y_d(\mathbf{x})$ equal or larger than l_c. This promotes the identification of a corrective GP model $y_d(\mathbf{x})$ that captures the long-term variations of the degradation process. In order to accelerate the training process, the training data is collected every 10 cycles.

The three test cases show that the accuracy of the surrogate models is affected by the similarity between the sources. The co-kriging model and the multi-output GP model have similar predictions in the presence of highly correlated sources (Fig. 4 (a)). Both models generate accurate predictions of the degradation profile of the primary cell A3 when the auxiliary cell is A4. The predictions of the models differ as the similarity between the sources drops (Fig. 4 (b) and Fig. 4 (c)). The multi-output model generates tighter confidence intervals than the co-kriging surrogate. The Supplementary Information contains plots of the predicted degradation profiles at 300 and 500 cycles. Furthermore, the document contains the RMSE and correlation values between the predicted and the denoised degradation profile of A3. This information is summarized in the box-plots in Fig. 4 (d). The co-kriging and multi-output models perform similarly in the first two test cases while the co-kriging model outperforms the multi-output model in the last test case.

In the test cases of dataset B, the primary cell is B1 and the auxiliary cells are B3, B2, and B4. This dataset is utilized to test the performance of the forecasting models in the presence of noisy data. The performance of the surrogate models is studying using high-fidelity information from the first 25, 50, 75, 100, 125, and 150 cycles.

The mean function of the GP model of $y_c(\mathbf{x})$ is a linear function with a prior distribution $\boldsymbol{\beta}_c \sim \mathcal{N}(\mathbf{b}_c, \mathbf{B}_c)$. The basis functions $\mathbf{h}_c(\mathbf{x})$, and the prior

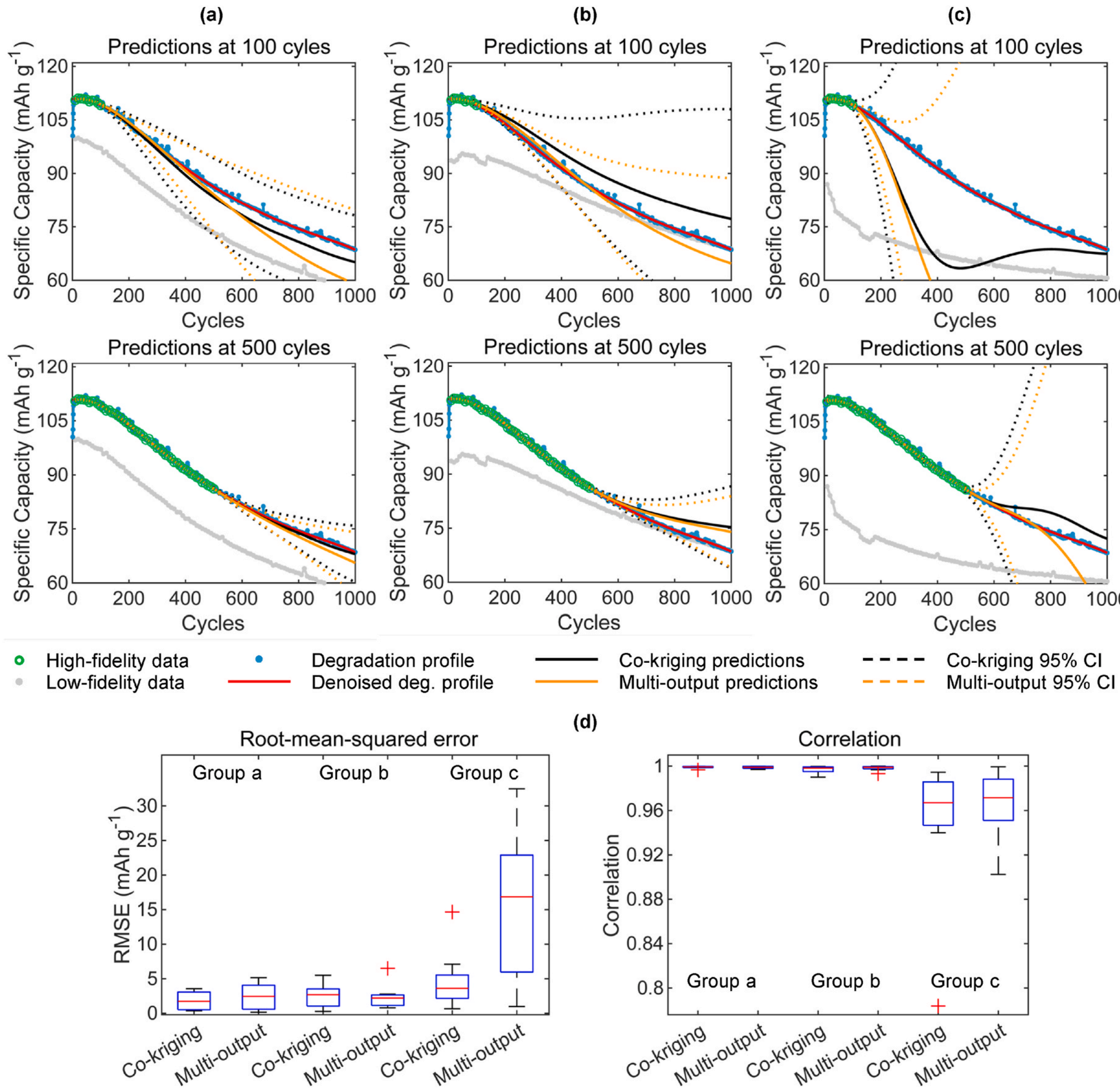

Fig. 4. Dataset A results. (a) Predicted degradation profile of A3 when the auxiliary cell is A4 at cycles 100 and 500. **(b)** and **(c)** show the predicted degradation profile when the auxiliary cells are A2 and A1, respectively. In these figures, the red curve is the denoised degradation profile of A3, the dotted lines represent the 95% confidence intervals generated by the co-kriging and multi-output models. **(d)** Box-plots of the RMSE and correlation values generated by the predictive models. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

parameters $\mathbf{b}_c$, and $\mathbf{B}_c$ are

$$\mathbf{h}_c(\mathbf{x}) = \begin{bmatrix} 1 \\ x \end{bmatrix},\ \mathbf{b}_c = \begin{bmatrix} y_0 \\ m_c \end{bmatrix},\ and\ \mathbf{B}_c = \begin{bmatrix} 0.5 & 0 \\ 0 & {m_c}^2 \end{bmatrix}, \tag{29}$$

where $y_0 = (\sum_{i=1}^{10} y_i)/10$, $m_c = (y_n - y_0)/x_n$, y_i is the capacity at the i-th cycle, and y_n is the capacity measured at the last cycle x_n. The corrective model $y_d(\mathbf{x})$ has a constant mean function with a prior distribution $\boldsymbol{\beta}_d \sim \mathcal{N}(\mathbf{b}_d, \mathbf{B}_d)$. The mean function of $y_d(\mathbf{x})$ is composed by

$$\mathbf{h}_d(\mathbf{x}) = 1,\ \mathbf{b}_d = 0, \quad \textbf{\textit{and}}\ \mathbf{B}_d = 1. \tag{30}$$

During training, the optimizer is required to find a length scale l_c of the low-fidelity source $y_c(\mathbf{x})$ larger than 20 cycles and a length scale l_d of the difference model $y_d(\mathbf{x})$ equal or larger than l_c in order to prevent overfitting.

The results in Fig. 5 shows once again that the accuracy of the surrogate models is affected by the similarity between the sources, and that the co-kriging and multi-output GP models have similar predictions when the sources have high similarity (Fig. 5 (a)). The performance indices show that the co-kriging surrogate outperforms the multi-output model in all test cases (Fig. 5 (d)). The RMSE and correlation values of the last test case show the benefits of employing a co-kriging model with informative mean functions. The auxiliary cell B4 registers measures of the capacity up to 132 cycles (Fig. 5 (c)). The informative mean function accounts for the missing information that if available could be used to predict the degradation of the primary cell B1 beyond the 132 cycles. The multi-output GP model does not have a mechanism to compensate the missed information. The Supplementary Information contains plots of the predicted degradation profiles at cycles 50 and 100 as well as the performance indices that generate the box-plots in Fig. 5 (d).

In the test cases of dataset C, the primary cell is C1 and the auxiliary cells are C2, C4, and C3. This dataset is utilized to test the performance of the models in the presence of noisy data and non-linear degradation profiles. To study the performance of the surrogate models, high-fidelity information from the first 25, 50, 75, 100, and 125 days is utilized. The co-kriging model is trained using the settings in Equation (29) and Equation (30). During training, the optimizer is required to find a length scale l_c of the low-fidelity source $y_c(\mathbf{x})$ larger than 20 days and a length

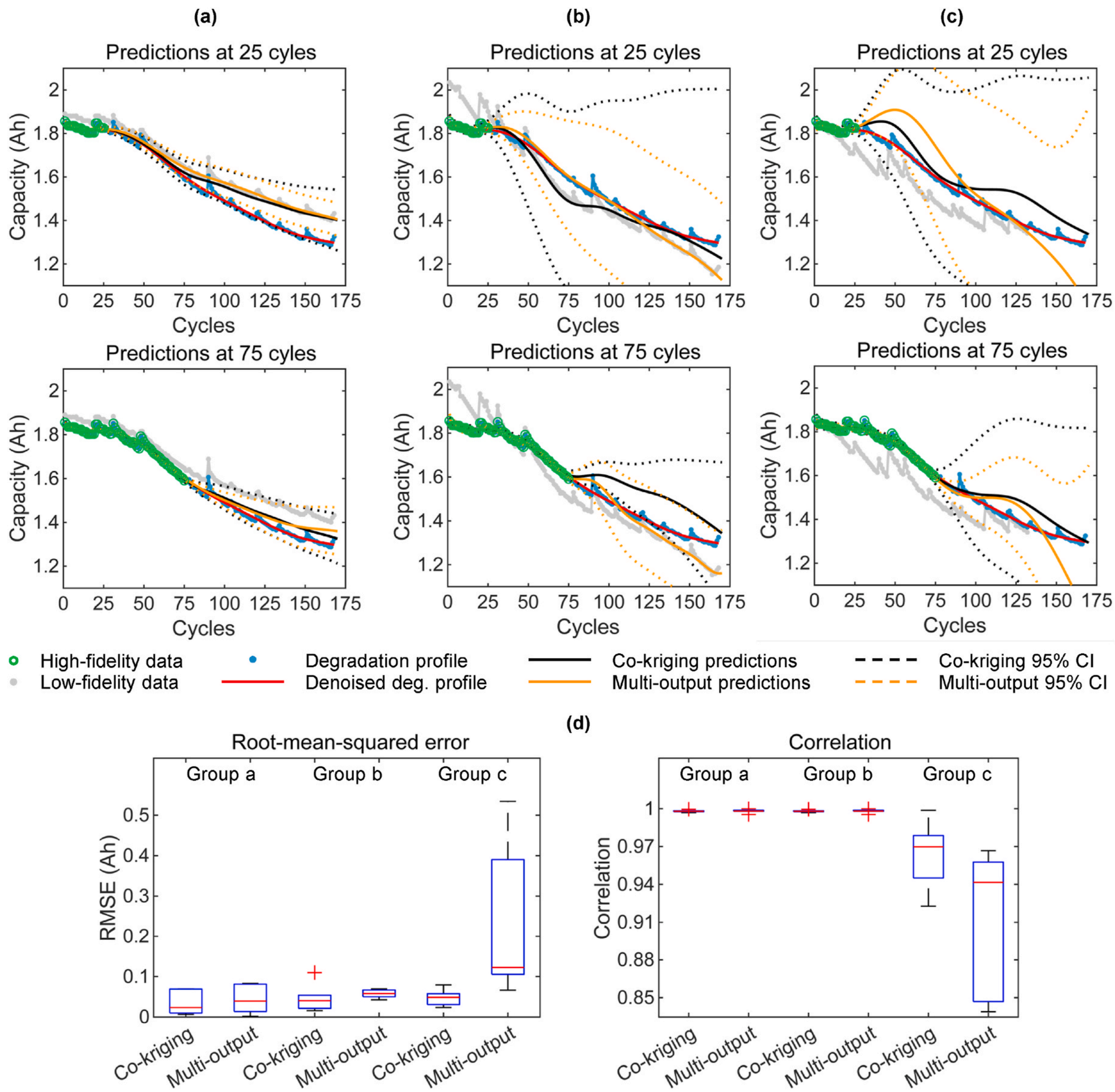

Fig. 5. Dataset B results. **(a)** Predicted degradation profile of B1 when the auxiliary cell is B4 at cycles 25 and 75. **(b)** and **(c)** show the predicted degradation profile when the auxiliary cells are B2 and B1, respectively. In these figures, the red curve is the denoised degradation profile of B1, the dotted lines represent the 95% confidence intervals generated by the co-kriging and multi-output models. **(d)** Box-plots of the RMSE and correlation values generated by the predictive models. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

scale l_d of the difference model $y_d(\mathbf{x})$ equal or larger than l_c in order to prevent overfitting.

In this dataset, the multi-output GP model tends to produce tight confidence intervals even if there is few high-fidelity data (Fig. 6 (a)) or low similarity between the sources (Fig. 6 (c)). The co-kriging model predicts more conservative confidence intervals. The co-kriging surrogate outperforms the multi-output model in all test cases (Fig. 6 (d)). The randomized nature of the cycling loading in the dataset is reflected in the spread of the box-plots of the RMSE, which indicates that in some instances the co-kriging and multi-output GP models do not accurately predict the degradation of the primary cell C1. The Supplementary Information contains plots of the predicted degradation profiles at days 50 and 100 as well as the performance indices that generate the box-plots in Fig. 6 (d).

3.2. Design optimization of cathode active materials for LIBs

This section employs Bayesian optimization to identify the Ni contents of cathode active materials $LiNi_xMn_{2-x}O_4$ that can produce cells with high initial specific capacity and\or large cycle life. The training data is extracted from the capacity degradation profiles illustrated in Fig. 2 (a). Given that the capacities of four of the five cells increase during the first cycles of the test, the initial specific capacity is defined as the average of the hundred largest values. The cycle life is defined as the number of cycles until 80% of the initial specific capacity is reached. Table 2 shows the initial specific capacity and cycle life of the tested cells. Since the specific capacity of the $LiNi_{1.0}Mn_{1.0}O_4$ cell does not reach an 80% degradation by the completion of the cycling test, a linear approximation with information from the last 500 cycles is used to estimate the cycle life of the cell.

The formulation of the optimization problem is

$$\begin{array}{ll} \text{Find} & x \in \mathbb{R} \\ \text{Maximize} & f_1(x): \ \text{Cycle life} \\ \text{Maximize} & f_2(x): \text{Initial specific capacity} \\ \text{Subject to}: & 0 \leq x \leq 1, \end{array} \tag{31}$$

where x is the content of Ni in the active material $LiNi_xMn_{2-x}O_4$.

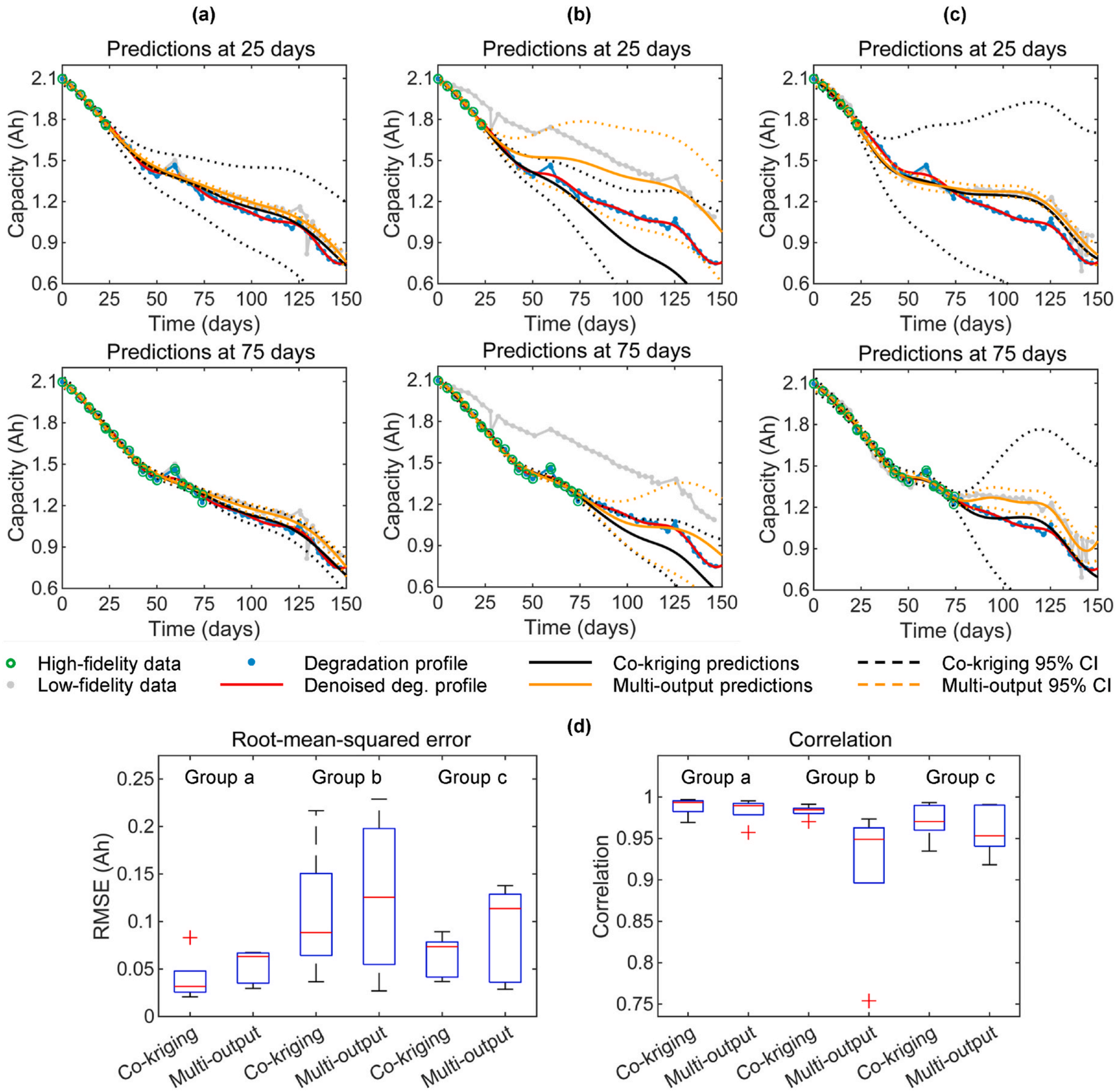

Fig. 6. Dataset C results. (a) Predicted degradation profile of C1 when the auxiliary cell is C2 at days 25 and 75. **(b)** and **(c)** show the predicted degradation profile when the auxiliary cells are C4 and C3, respectively. In these figures, the red curve is the denoised degradation profile of C1, the dotted lines represent the 95% confidence intervals generated by the co-kriging and multi-output models. **(d)** Box-plots of the RMSE and correlation values generated by the predictive models. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 2
Training data. Initial specific capacity and cycle life of the five CR2032 coin cells.

Ni content	0.0	0.4	0.5	0.6	1.0
Initial capacity (mAh g^{-1})	89.3	95.4	111.3	99.9	59.8
Cycle life (cycles)	221	686	446	374	3281

The GP regression models of the specific capacity and cycle life employ a squared-exponential covariance function and a constant mean function (Fig. 7 (a)). The predictions of these models are utilized in the formulation of the EEI and goal-based EEI functions. The goal-based EEI function incorporates the preferences of the designer in the optimization process by prescribing goals that guide the optimization towards zones of interest. In order to promote a uniform expansion of the Pareto front, five well-spaced design goals are defined (Table 3 and Fig. 7 (c)).

Fig. 7 (b) shows the acquisition landscapes of the EEI function and the goal-based EEI function that employs the fifth design goal. The maximum of the EEI function is a design that improves the overall quality of the current Pareto front while the maximum of the goal-based EEI function is a design that can expand the surroundings of the prescribed goal, i.e., the upper left corner of the Pareto front.

Fig. 7 (c) shows the mean predictions of the GP regression models for different Ni compositions, the design that maximizes the EEI function, the prescribed goals, and the designs that maximize the goal-based EEI function. The current Pareto designs are the cells with Ni contents of 0.5, 0.4, and 1.0. The purpose of the EEI and goal-based EEI acquisition functions is to identify new designs that can improve the current Pareto front if sampled (tested).

The optimal EEI design has a Ni content of 0.81. This design is selected by the optimizer since it can improve the overall quality of the current Pareto front by filling the gap between the cells with Ni contents of 0.4 and 1.0. Instead of improving the whole Pareto front, maximization of the goal-based EEI function leads to designs that improve zones of the Pareto front that surround the prescribed goals. The optimal goal-based EEI designs have Ni contents of 0.94, 0.90, 0.83, 0.75, and 0.44. By simultaneously testing these designs, several areas of the current

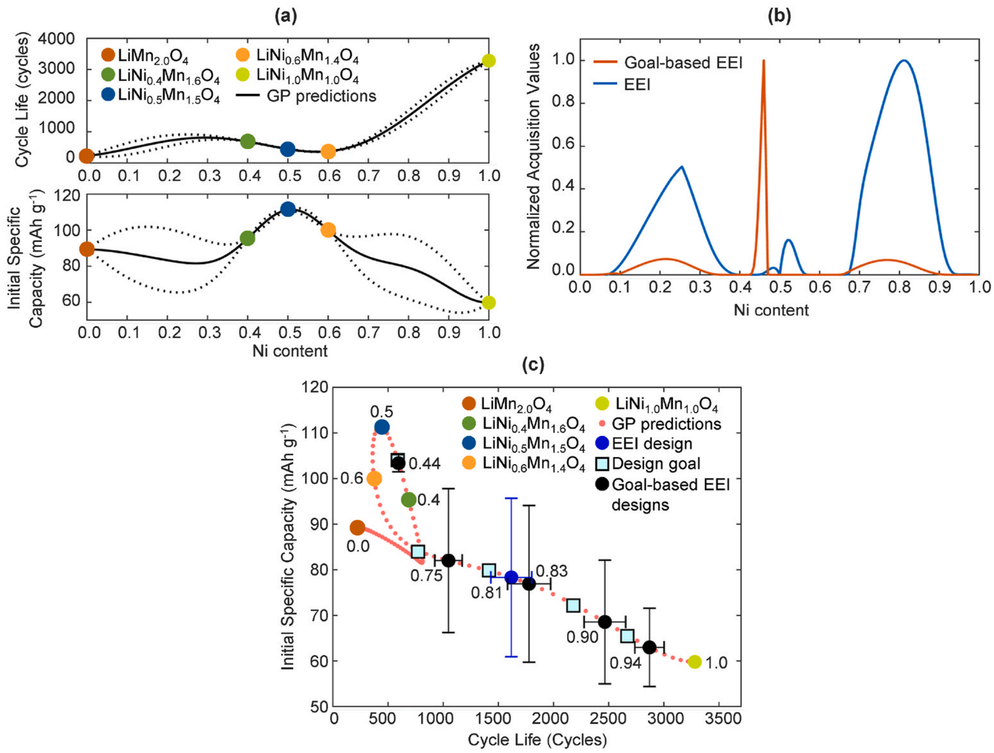

Fig. 7. Bayesian optimization of cathode active materials. (a) GP regression models of the initial specific capacity and cycle life. The dotted lines denote the 68% confidence interval. (b) Acquisition landscapes of the EEI function and the goal-based EEI function that employs the fifth design goal (Table 3). (c) Design map generated by data from the optimization process. The map contains the training data, the GP predictions of the designs selected by the optimizer and their 68% confidence bounds, the GP predictions of other designs, and the most relevant Ni contents.

Table 3
Goal-based EEI. Prescribed goals to promote a uniform expansion of the current Pareto front.

Prescribed goal	1	2	3	4	5
Initial capacity (mAh g^{-1})	65.5	72.2	79.9	84.0	104.1
Cycle life (cycles)	2669	2179	1414	771	588

Pareto front can be improved in a single iteration. In the case of large uncertainty in the GP predictions, the identified designs depart from the prescribed goals to balance exploration of the design space and exploitation of the surrogate models.

Table 4 contains the Ni contents identified by the EEI and goal based EEI functions and the 95% confidence interval of the initial specific capacity and cycle life predicted by the GP models. The goal-based EEI 5 design exploits the upper left corner of the Pareto front while the other goal-based EEI designs explore the current Pareto front.

The information generated by the Bayesian optimization approach can assist the design of future experiments. A design map similar to Fig. 7 (c) can be used to decide the next batch of active materials to be tested. For example, if the goal of the next batch of experiments is to improve the diversity of the current Pareto front, i.e., exploration, Ni contents of 0.7, 0.8, and 0.9 can be tested. If the designer wants to focus the search over cells with specific characteristics, e.g., large specific capacity, a set of prescribed goals along the upper left corner of the Pareto front can be defined.

Table 4
Bayesian optimization results. Ni contents of the optimal designs and their GP predictions. The predictions correspond to the 95% confidence interval.

Optimization strategy	Ni content	Initial capacity (mAh g^{-1})	Cycle life (cycles)
EEI	0.81	78.3 ± 34.8	1617 ± 372
Goal-based EEI 1	0.94	63.0 ± 17.2	2870 ± 266
Goal-based EEI 2	0.90	68.6 ± 27.2	2465 ± 378
Goal-based EEI 3	0.83	76.9 ± 34.4	1778 ± 392
Goal-based EEI 4	0.75	82.0 ± 31.6	1048 ± 248
Goal-based EEI 5	0.44	103.4 ± 3.8	595 ± 12

4. Conclusion

This study presents the use of two GP-based techniques, i.e., co-kriging surrogates and Bayesian optimization, in the prediction of the capacity degradation of LIBs under cycling loads and in the design of new cathode active materials that can produce cells with high initial specific capacity and large cycle life. Three datasets are employed in this work. The first dataset comes from the cycling performance test of five CR2032 cells with $LiNi_xMn_{2-x}O_4$ cathodes, where each cell has a Ni content of 0.0, 0.4, 0.5, 0.6, and 1.0 in their cathode active material. The last two datasets are the capacity degradation profiles of eight commercial 18650 batteries taken from the open-source repository of the NASA Ames Prognostics Center of Excellence [31,32].

The three datasets are utilized to study and compare the performance of the co-kriging surrogate and a multi-output GP regression model [19]. These forecasting models combine two sources of information: the early cycling data of a cell (referred as the primary cell) and the complete cycling information of a related cell (referred as the auxiliary cell) in order to predict the capacity of the primary cell in the remaining cycles. The results show that the co-kriging surrogate and the multi-output GP regression model have accurate predictions when the sources have a high degree of similarity. In the case of intermediate and low similarity, the multi-output GP model tends to overfit the data whereas the co-kriging surrogate displays a robust behavior. The performance metrics (RMSE and correlation coefficient) indicate that the co-kriging surrogate outperforms the multi-output GP model in most of the test cases. The results also show the ability of the models to filter the noise in the training data.

In the design of cathode active materials, the data from the five CR2032 cells is utilized. Two GP regression models are trained to predict the initial specific capacity and the cycle life of the LIB cell as a function of the Ni content in the cathode active material. The EEI and goal-based EEI acquisition functions are compared. The results show that the EEI identifies designs that can improve the overall quality of the current Pareto front whereas the goal-based EEI function utilizes a prescribed design goal to identify designs that improve specific zones of the Pareto front. The utilization of well-spread design goals enables parallel optimization and lead to designs that can uniformly expand the current Pareto front. The results suggest that a Ni content of 0.44 produces cells with an initial specific capacity of 103.4 ± 3.8 mAh g^{-1} and cycle life of 595 ± 12 cycles, and that Ni contents of 0.7, 0.8, and 0.9 can improve the diversity of the current Pareto front.

In this study, the scaling factor is set to $\rho = 1$, in order to capture the overall correlation between the degradation profiles of the primary and auxiliary cells. This setting leads to better results than the inclusion of ρ in the optimization of the hyperparameters of $y_d(\mathbf{x})$. Ongoing work involves the development of a co-kriging surrogate whose scaling parameter ρ has a prior distribution that captures a prior knowledge about the similarity between the sources. The new formulation eliminates the closure property of GPs but can enhance the robustness of the co-kriging formulation. Another ongoing effort is the development of a co-kriging surrogate capable of handling outliers in the experimental data. It is important to notice that the low-fidelity information is not limited to the testing information of a related cell. An interesting alternative is the use of data generated by numerical simulations as low-fidelity information. The combination of experimental and numerical data can accelerate the design of new active materials.

Data availability

The data that support the findings of this study are available on request from the corresponding author.

CRediT authorship contribution statement

Homero Valladares: Methodology, Writing – original draft, Writing – review & editing, Software, Formal analysis. **Tianyi Li:** Investigation, Validation, Writing – original draft. **Likun Zhu:** Conceptualization, Investigation, Validation, Writing – review & editing. **Hazim El-Mounayri:** Conceptualization, Investigation, Writing – review & editing. **Ahmed M. Hashem:** Investigation, Validation, Writing – review & editing. **Ashraf E. Abdel-Ghany:** Investigation, Validation, Writing – review & editing. **Andres Tovar:** Conceptualization, Writing – original draft, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jpowsour.2022.231026.

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